

Problem 13. Find all ordered pairs (a, b) of positive integers for which the numbers

$$\frac{a^3b - 1}{a + 1} \quad \text{and} \quad \frac{b^3a + 1}{b - 1}$$

are both integers.

Problem 14. Let Γ be the circumcircle of an acute-angled triangle ABC . The perpendicular to AB from C meets AB at D and Γ again at E . The bisector of angle C meets AB at F and Γ again at G . The line GD meets Γ again at H and the line HF meets Γ again at I . Prove that $AI = EB$.

Problem 15. What is the least possible number of cells that can be marked on an $n \times n$ board such that for each $m > n/2$ both diagonals of any $m \times m$ sub-board contain a marked cell?

Problem 16. Let $\mathbb{R}$ denote the set of real numbers. Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x^2) + f(xy) = f(x)f(y) + yf(x) + xf(x+y), \quad \text{for all } x, y \in \mathbb{R}.$$